Skill Builder Pack 4 – Analysing Blended Data in Excel

Purpose

Once data has been blended into a single dataset, the next step in the analysis process is to extract useful insights.

Analysis means asking questions about the data and using Excel functions to answer them.

In many organisations, once datasets have been combined, analysts need to:

- Understand patterns (e.g. attendance trends)
- Identify risks (e.g. low engagement students)
- Summarise information (e.g. averages, totals)

Excel provides a range of simple functions that allow this to be done efficiently.

In this skill builder, you will use:

- Filters
- IF and IFS
- SUM and AVERAGE
- COUNT, COUNTIF, COUNTIFS
- SUMIF, SUMIFS
- AVERAGEIF, AVERAGEIFS

Learning Outcomes

By the end of this skill builder you should be able to:

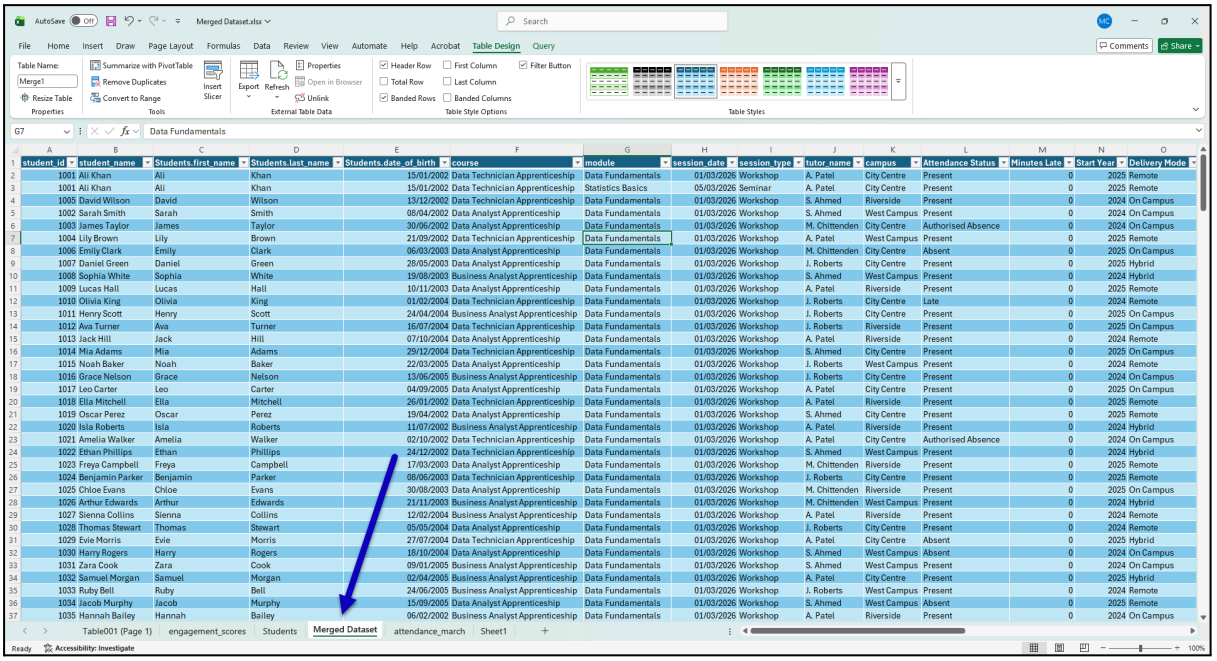
- Ask meaningful questions about a dataset
- Use Excel functions to answer those questions
- Apply conditional logic using IF and IFS
- Perform calculations using SUM, COUNT and AVERAGE functions
- Use conditional functions such as SUMIF and COUNTIFS
- Interpret results to support decision making

Starting Point

You should now have a **single merged dataset** containing:

- Student details (name, course, email)
- Attendance data (status, minutes late, session type)
- Engagement data (scores and risk band)
- Withdrawal information

For this use the Merged Dataset.xslx file in the Module 4 folder and choose the **Merged Dataset** tab.



student_id	student_name	Students.first_name	Students.last_name	Students.date_of_birth	course	module	session_date	session_type	tutor_name	campus	Attendance Status	Minutes Late	Start Year	Delivery Mode
1001	Ali Khan	Ali	Khan	15/01/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0	2025	Remote
1001	Ali Khan	Ali	Khan	15/01/2002	Data Technician Apprenticeship	Statistics Basics	05/03/2026	Seminar	A. Patel	City Centre	Present	0	2025	Remote
1005	David Wilson	David	Wilson	13/12/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	Riverside	Present	0	2024	On Campus
1002	Sarah Smith	Sarah	Smith	08/04/2002	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	On Campus
1003	James Taylor	James	Taylor	30/06/2002	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Authorised Absence	0	2024	On Campus
1004	Lily Brown	Lily	Brown	21/09/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	West Campus	Present	0	2025	Remote
1006	Emily Clark	Emily	Clark	06/03/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Absent	0	2025	On Campus
1007	Daniel Green	Daniel	Green	28/05/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2025	Hybrid
1008	Sophia White	Sophia	White	19/08/2003	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	Hybrid
1009	Lucas Hall	Lucas	Hall	10/11/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2025	Remote
1010	Olivia King	Olivia	King	01/02/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Late	0	2024	Remote
1011	Henry Scott	Henry	Scott	24/04/2004	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2025	On Campus
1012	Ava Turner	Ava	Turner	16/07/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	Riverside	Present	0	2025	On Campus
1013	Jack Hill	Jack	Hill	07/10/2004	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	Remote
1014	Mia Adams	Mia	Adams	29/12/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Present	0	2025	On Campus
1015	Noah Baker	Noah	Baker	22/03/2005	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Present	0	2024	Remote
1016	Grace Nelson	Grace	Nelson	13/06/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2024	On Campus
1017	Leo Carter	Leo	Carter	04/09/2005	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0	2025	On Campus
1018	Ella Mitchell	Ella	Mitchell	26/01/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2025	Remote
1019	Oscar Perez	Oscar	Perez	19/04/2002	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Present	0	2025	Remote
1020	Isla Roberts	Isla	Roberts	11/07/2002	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	Hybrid
1021	Amelia Walker	Amelia	Walker	02/10/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Authorised Absence	0	2024	On Campus
1022	Ethan Phillips	Ethan	Phillips	24/12/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	Hybrid
1023	Freya Campbell	Freya	Campbell	17/03/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	Riverside	Present	0	2025	Remote
1024	Benjamin Parker	Benjamin	Parker	08/06/2003	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2025	Remote
1025	Chloe Evans	Chloe	Evans	30/08/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	Riverside	Present	0	2025	On Campus
1026	Arthur Edwards	Arthur	Edwards	21/11/2003	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	West Campus	Present	0	2024	Hybrid
1027	Sienna Collins	Sienna	Collins	12/02/2004	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	Remote
1028	Thomas Stewart	Thomas	Stewart	05/05/2004	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2024	Remote
1029	Evie Morris	Evie	Morris	27/07/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Absent	0	2025	Hybrid
1030	Harry Rogers	Harry	Rogers	18/10/2004	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Absent	0	2024	On Campus
1031	Zara Cook	Zara	Cook	09/01/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	On Campus
1032	Samuel Morgan	Samuel	Morgan	02/04/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0	2025	Hybrid
1033	Ruby Bell	Ruby	Bell	24/06/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Present	0	2024	Remote
1034	Jacob Murphy	Jacob	Murphy	15/09/2005	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Absent	0	2025	Remote
1035	Hannah Bailey	Hannah	Bailey	06/02/2002	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	On Campus

As you can see it is coloured differently to stand out.

Each row represents a **student session record**.

This means:

- A student may appear multiple times
- Each row represents one session (e.g. workshop, seminar, lab)

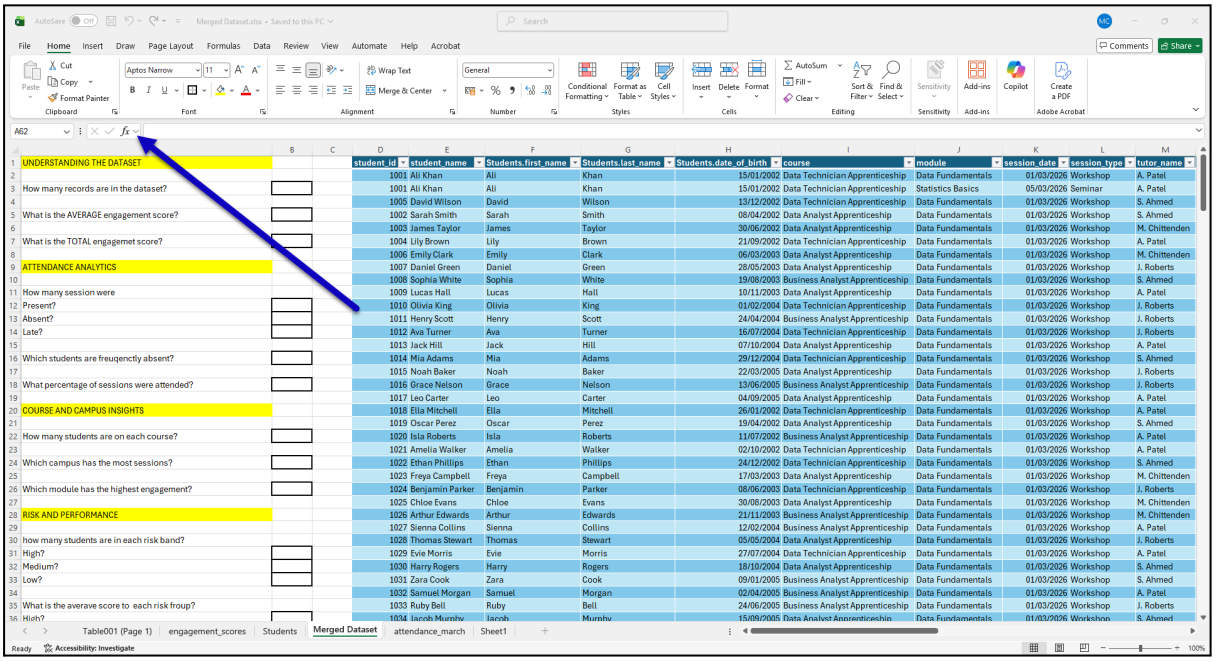
Key Idea

We are now moving from data preparation → data analysis.

Instead of combining data, we are now asking:

“What is this data telling us?”

First of all though, remember, this little ***fx*** here is your performance enhancing supplement - it’s going to do everything it can to make it easy for you. It’s the dictionary and wizard for **every** function in Excel that you will nee.



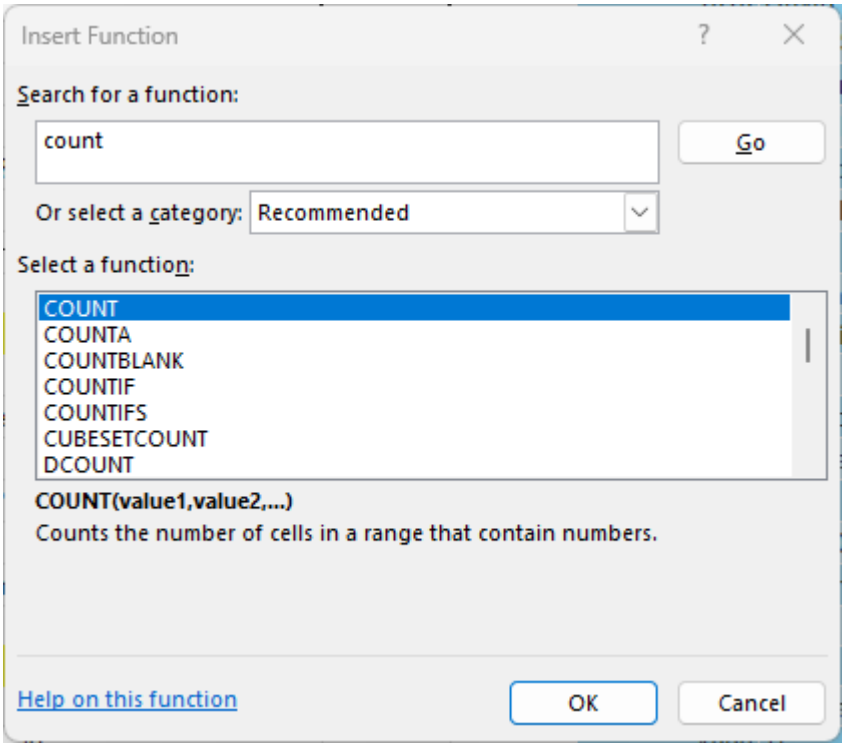
Questions We Want to Answer (and How We Answer Them)

1. Understanding the Dataset

How many records are in the dataset?

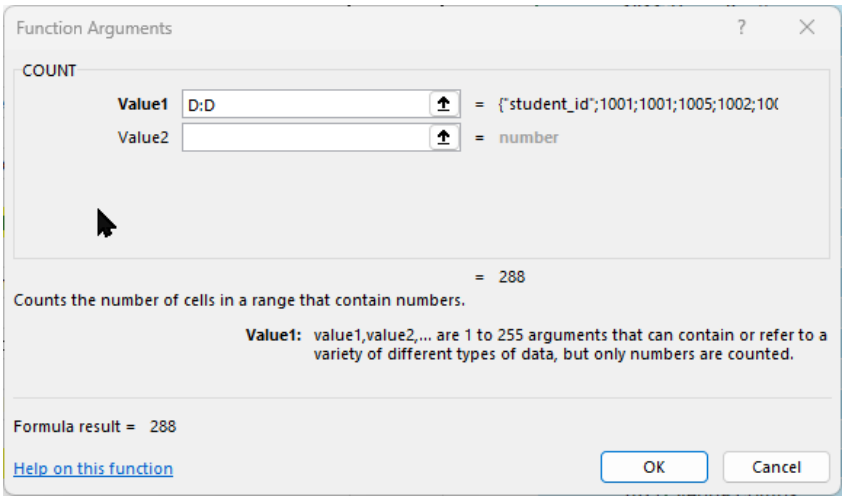
So this is a straightforward COUNT. This function returns a count of all rows in the dataset.

Choose cell **B3** and then click on ***fx*** and type COUNT



Click **OK**

Select the D column which contains **student_id**



Click **OK** to execute the formula.

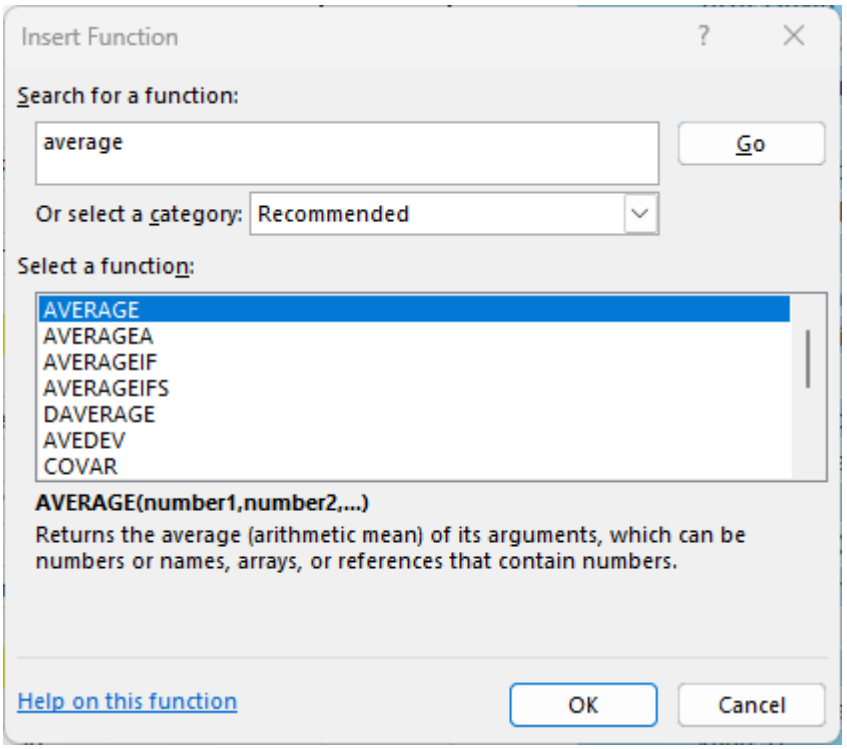
The answer to this is: **the total number of session records in the dataset** which should be **288.**

Notice it tells you this in a preview at the bottom as well

What is the average engagement score?

Now we will use AVERAGE. This function returns the average.mean score of all data selected/

Choose cell **B5** and then click on ***fx*** and type AVERAGE



Click **OK**.

Select all of column **T** - Engagement Score Weekly and click **OK**

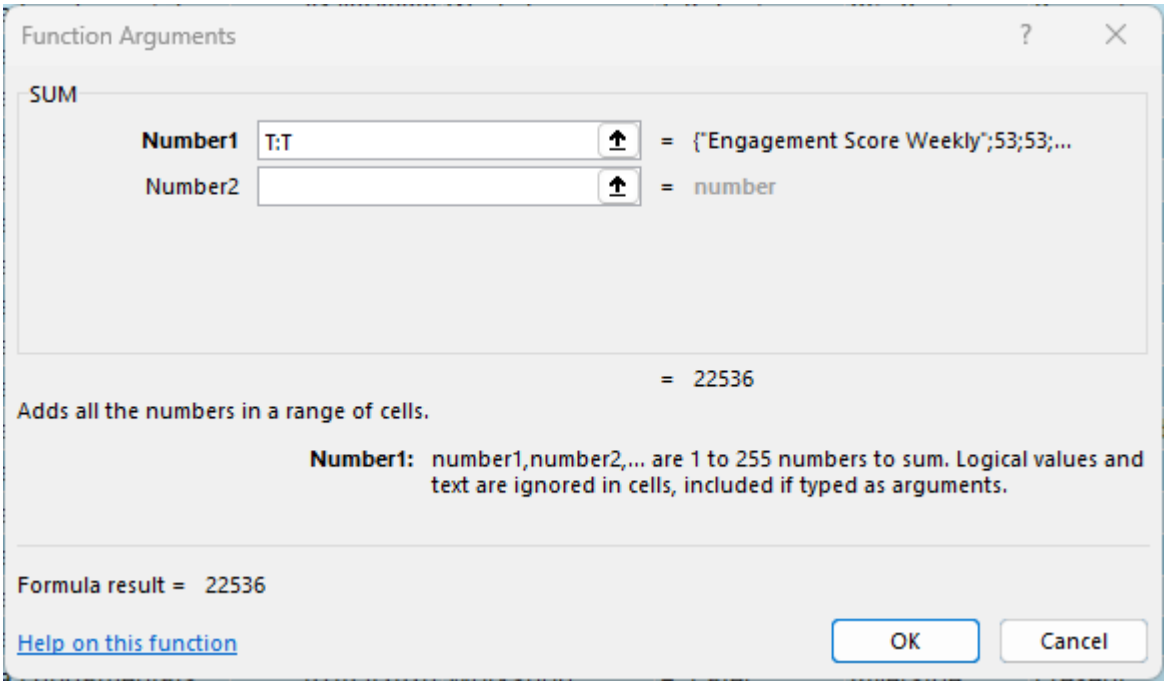
So the answer to this is: **the typical engagement level across all students** which should be 78.25

What is the total engagement score?

This is one of the common functions you probably use all the time - a straightforward SUM.

Select cell **B7** and then go to fx and type SUM and then click OK

Select column T again and click OK



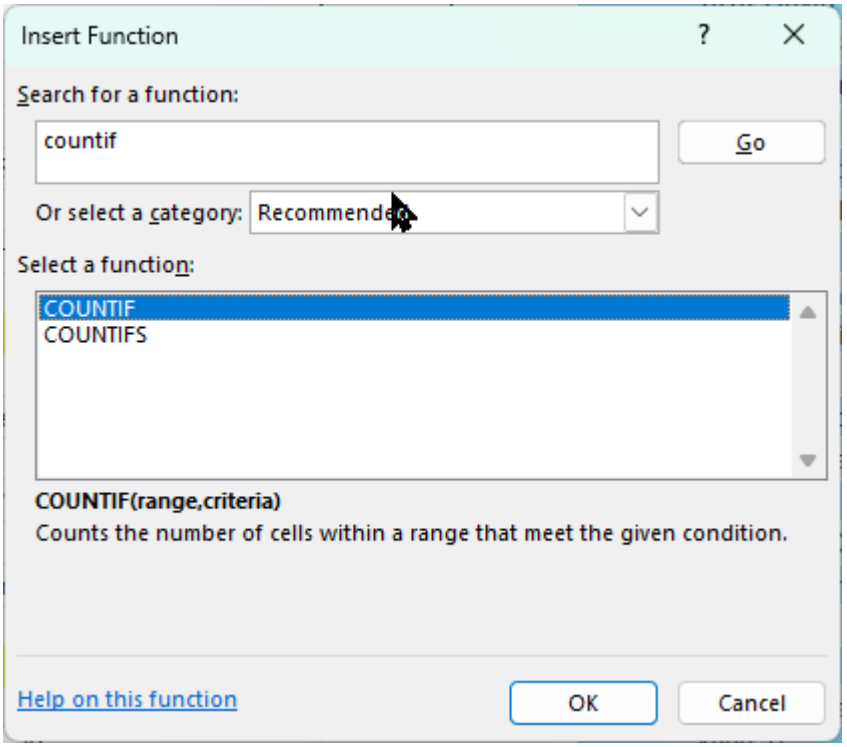
So the answer to this is: **the overall engagement across all sessions** which is 22, 536.

2. Attendance Analysis

How many sessions were Present, Absent, or Late?

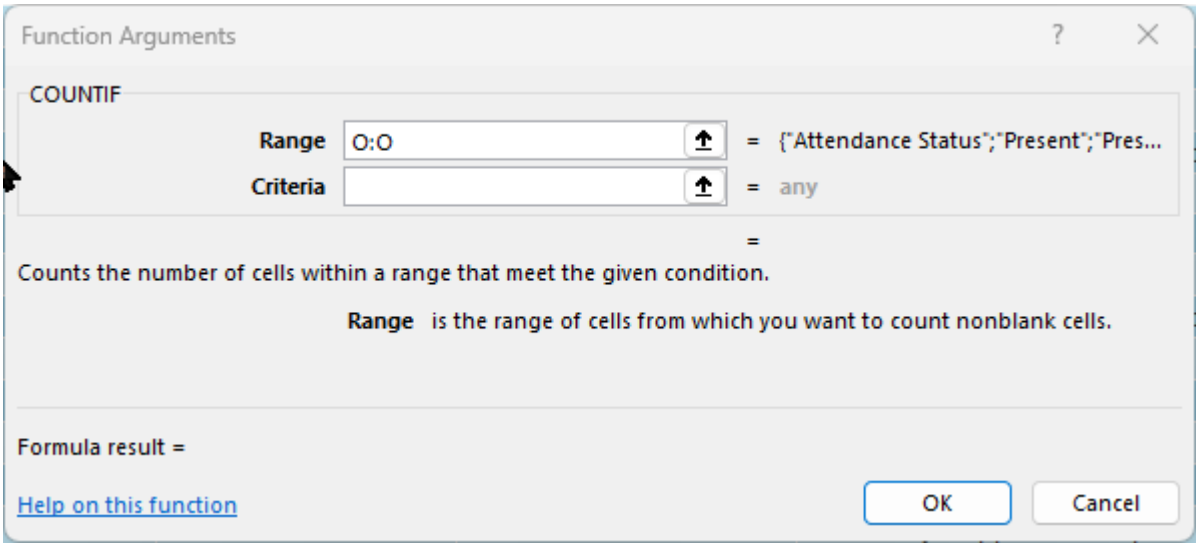
For thi we will use COUNTIF. You used COUNT previously, this is a more sophisticated and filtered version

Select cell B12 and then fx and search for COUNTIF (you might need to click on **Go** to filter the list)

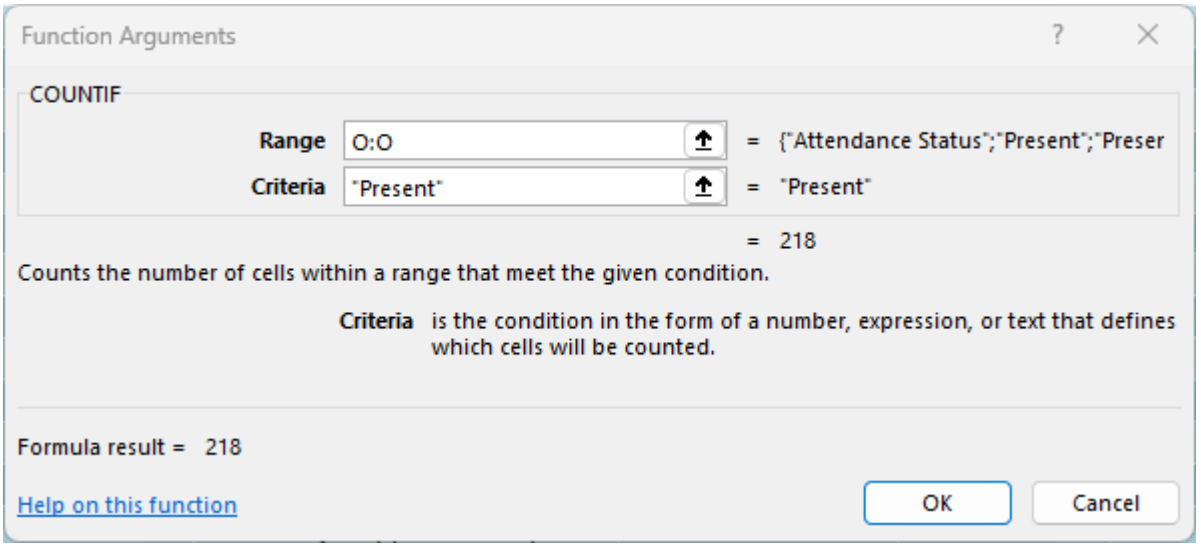


Once you have located it, click on **OK.**

So first we put in the range we want to check, in this case column O which contains the attendance data.



In the Criteria box we say what the filter it, in this first case it is Present



When we click on OK it will count only the data that says Present in this column and it should be 218.

Repeat for the other two absence categories (Absent and Late)

You should end up with this:

How many session were	
Present?	218
Absent?	65
Late?	5

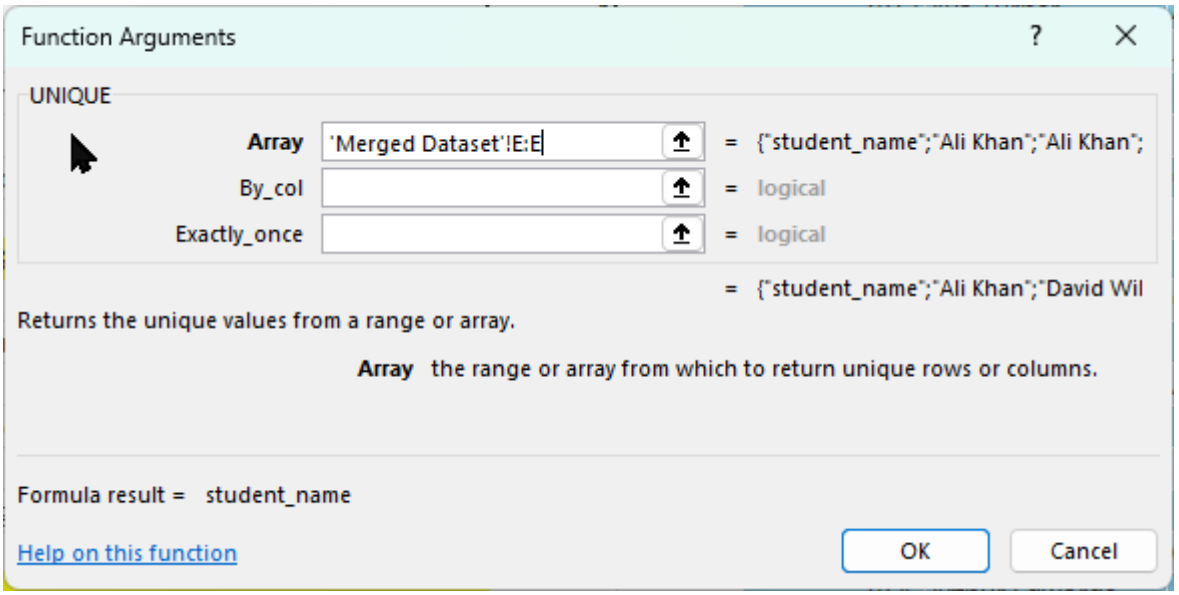
How could you check the accuracy of this figure, given something you previously calculated?

Which students are frequently absent?

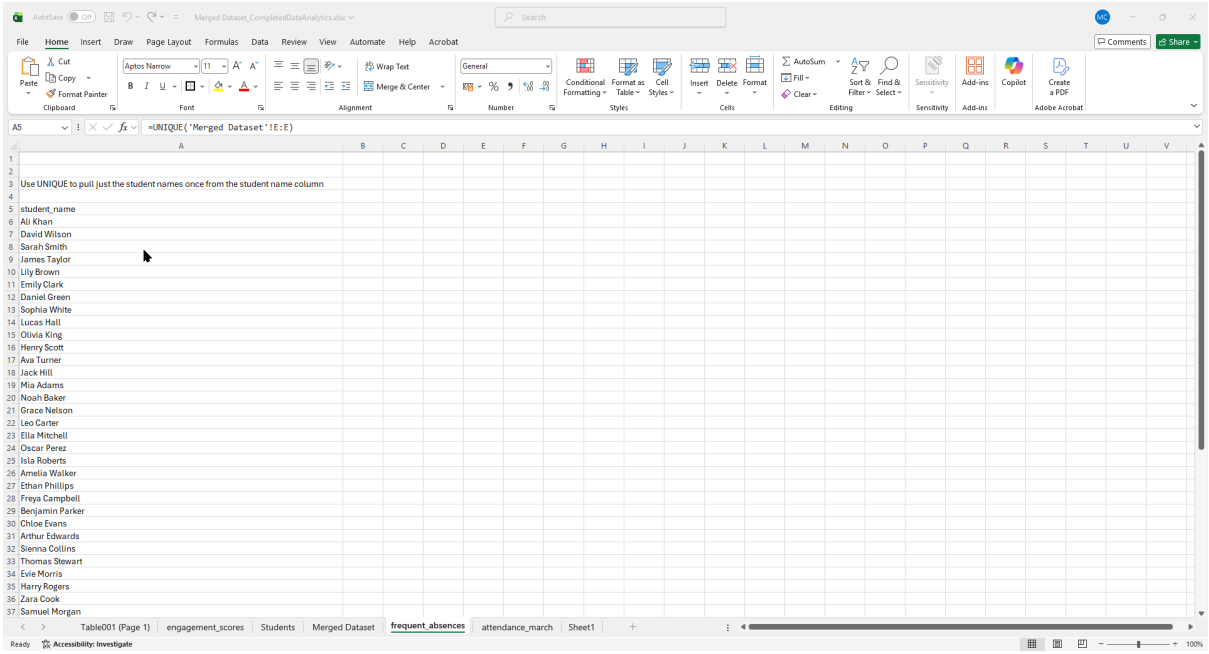
This one will probably be best worked with the **frequent_absences** spreadsheet.

Clearly many students can attend many courses so the first thing we want is a list of unique students and Excel has a function for that, called UNIQUE

We need to keep this nice and simple though - use fx again



This will end up giving you this list



These are all the unique learners appearing in the dataset.

Now we can use COUNTIFS if see how many times each has been absent.

COUNTIF and COUNTIFS are both used to count how many rows meet certain conditions in a dataset, but they are used in slightly different ways.

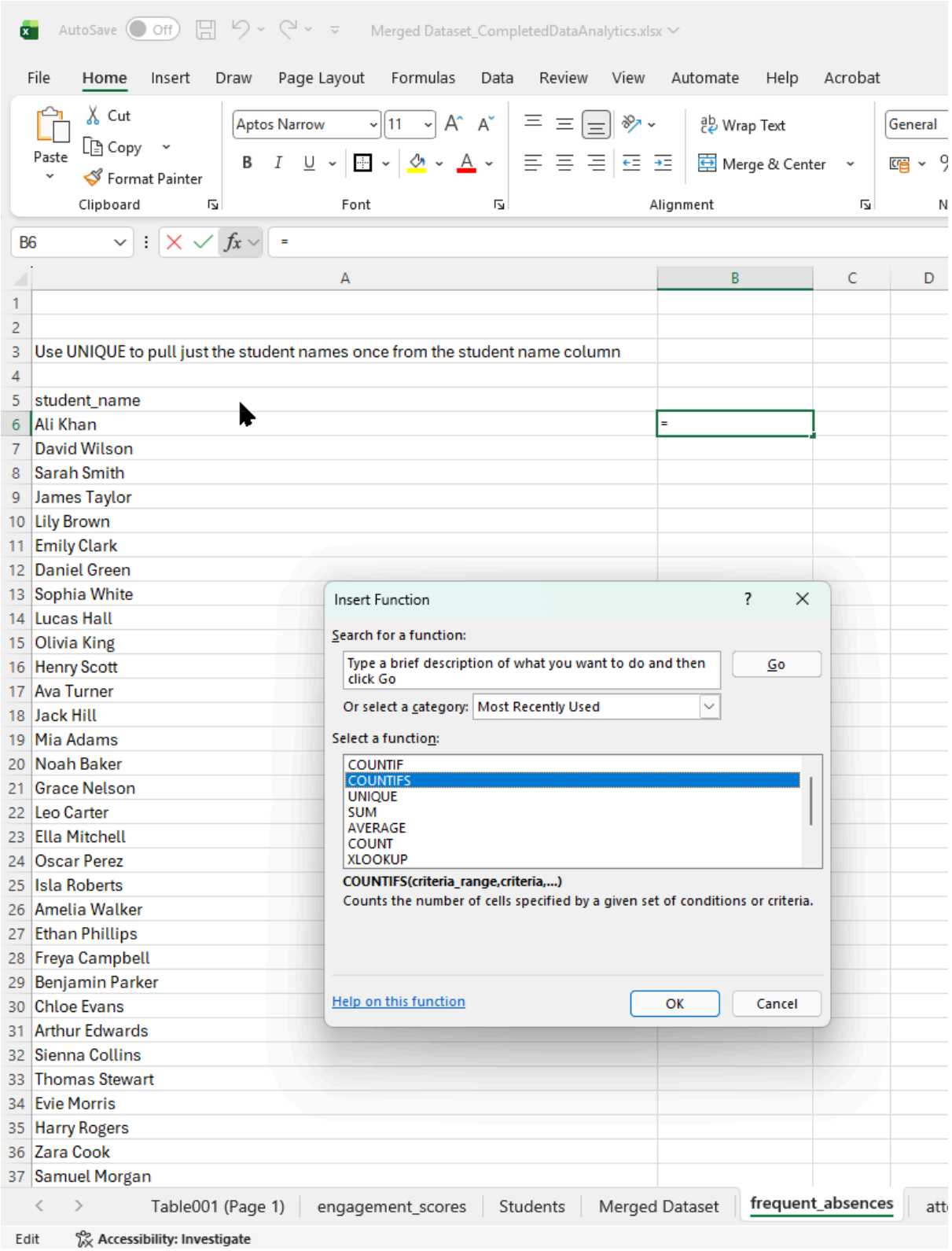
COUNTIF is used when you only have one condition to check. For example, if you want to count how many times students were absent, you can use COUNTIF to look through the Attendance Status column and count every time “Absent” appears. This is useful for simple questions where you are only checking one thing.

COUNTIFS is used when you need to check multiple conditions at the same time. For example, if you want to know how many students were both absent and in the high-risk category, COUNTIFS allows you to apply both of these conditions together. It will only count rows where all conditions are true.

The key difference is that COUNTIF works with one condition, while COUNTIFS works with two or more conditions. A simple way to think about it is that COUNTIF answers “how many meet this rule?”, whereas COUNTIFS answers “how many meet this rule and this rule at the same time?”.

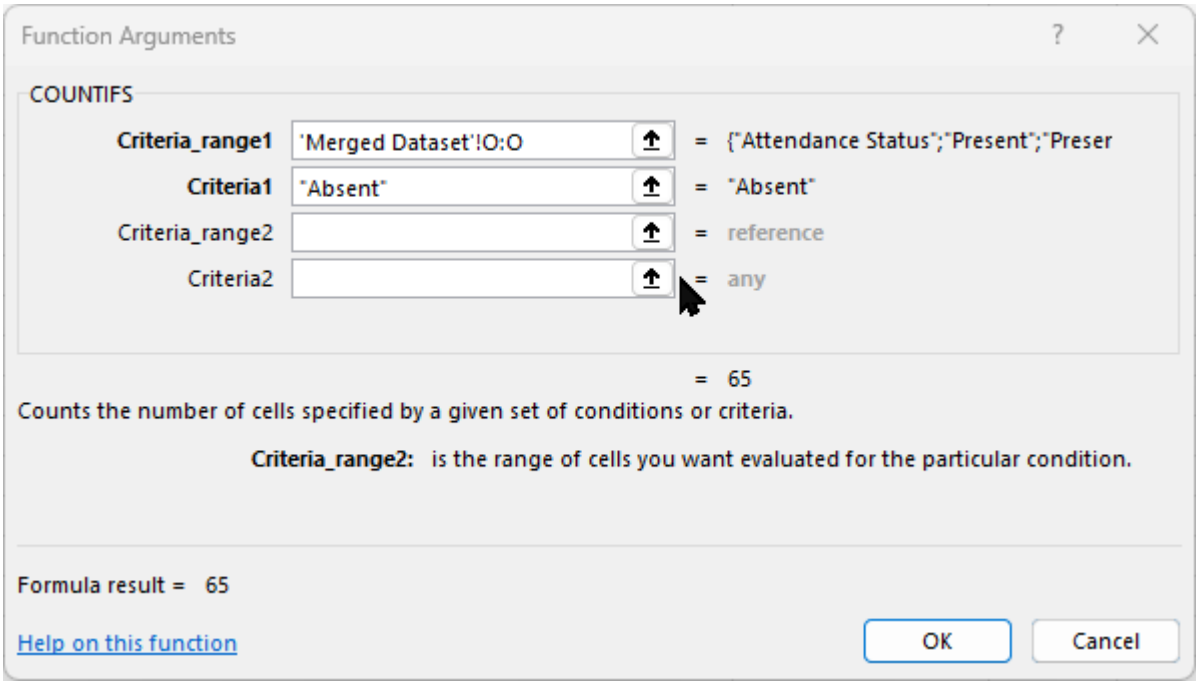
This can get complicated so take it slow.

Select a cell next to of your unique learners and use the fx method to search for COUNTIFS

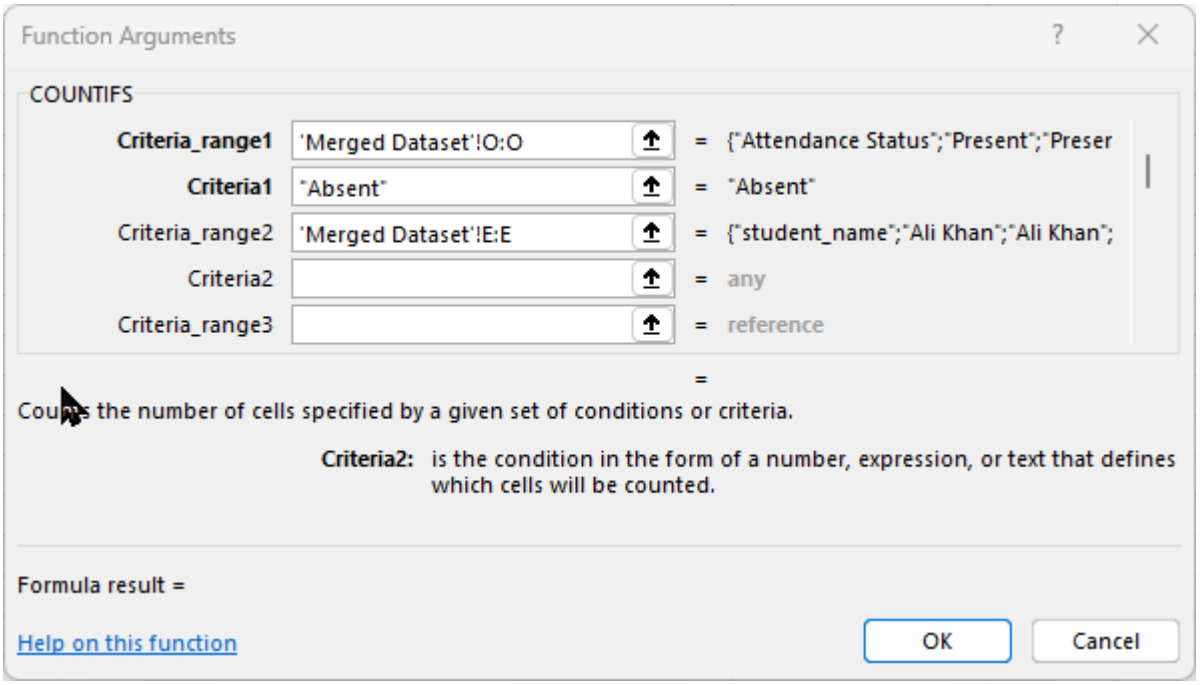


Click **OK**

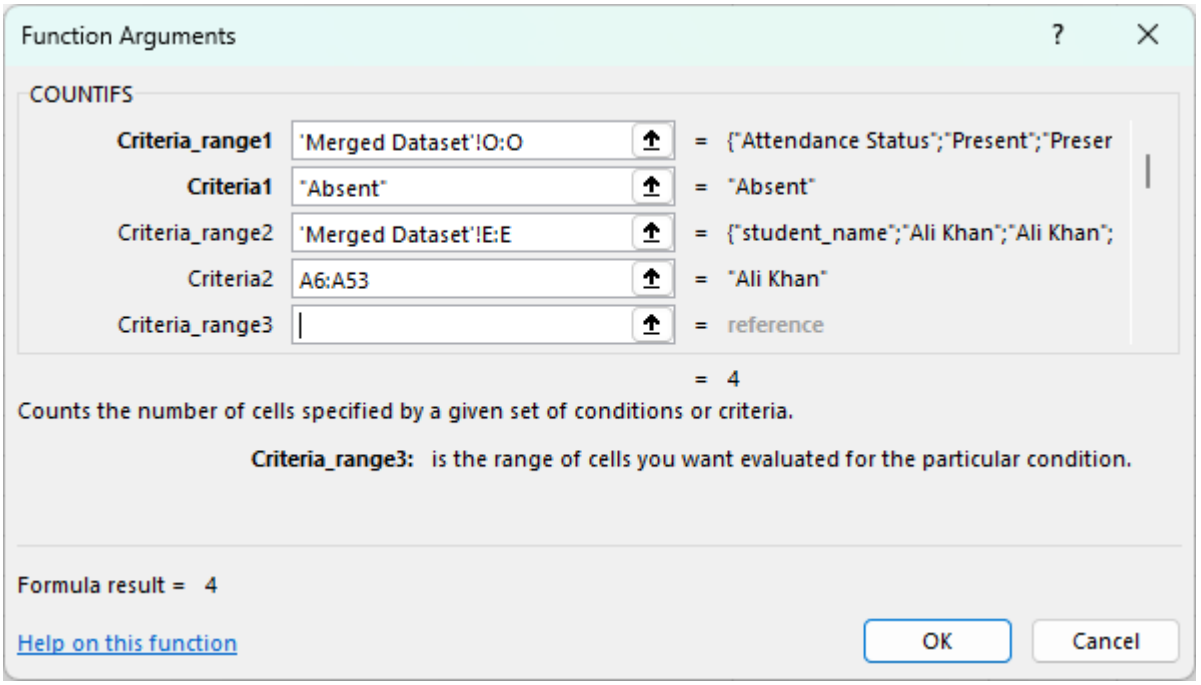
For **Criteria_range1** select the column containing the attendance data which is on the other sheet **Merged Dataset** (column O) and then for **Criteria1** put what you want to filter by, in this case classes where the learner was **Absent**



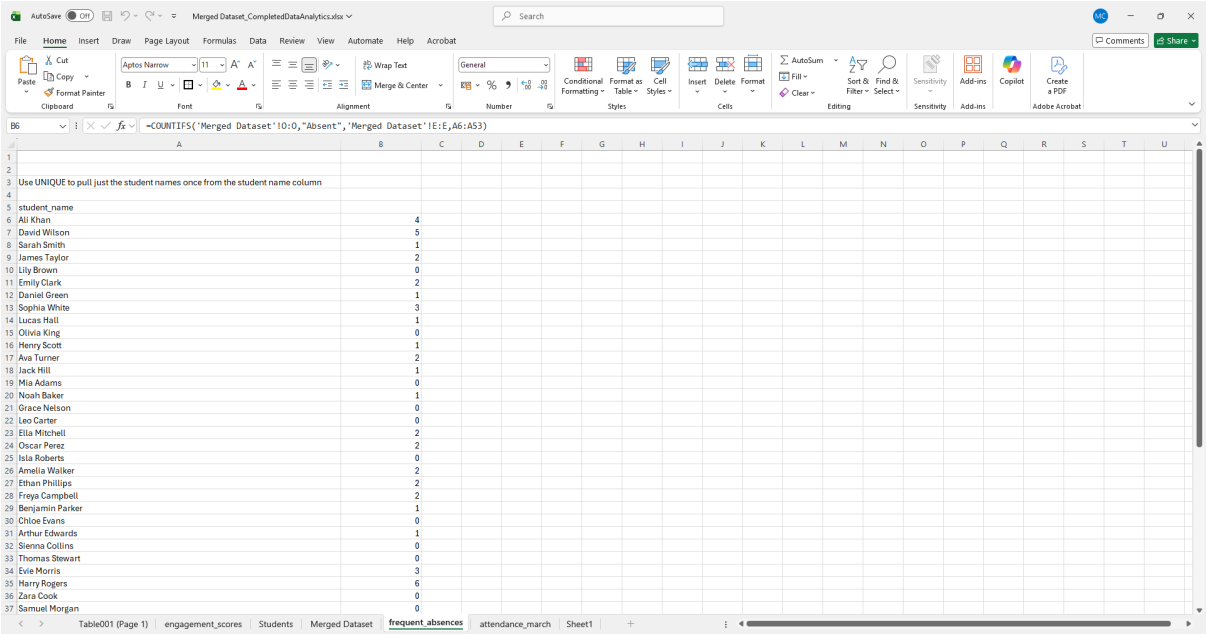
Then we want to filter by **name** so in the **Criteria_range2** pick the name column on the merged dataset (column E).



Now we get clever and because we already selected a unique lists of learners, our **Criteria2** is based upon that list.

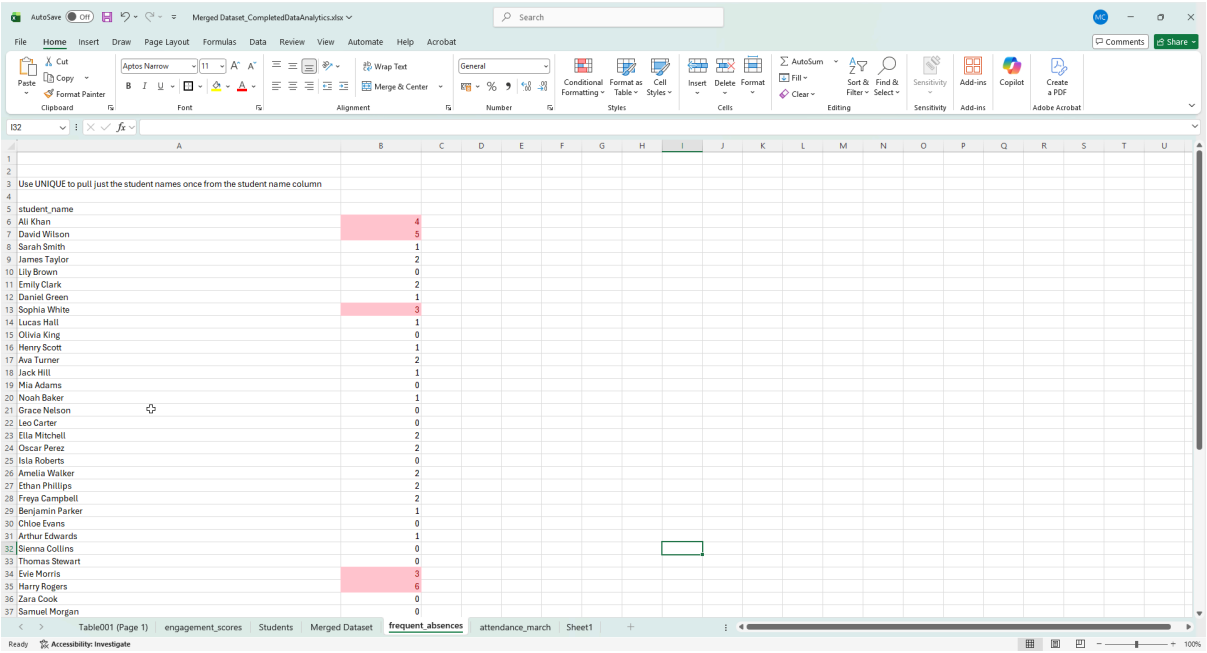


If you now click on **OK** that will give the count of how many time **Absent** has appeared against a learners name in our merged dataset.



student_name	
Ali Khan	4
David Wilson	5
Sarah Smith	1
James Taylor	2
Lily Brown	0
Emily Clark	2
Daniel Green	1
Sophia White	3
Lucas Hall	1
Olivia King	0
Henry Scott	1
Ava Turner	2
Jack Hill	1
Mia Adams	0
Noah Baker	1
Grace Nelson	0
Leo Carter	0
Ella Mitchell	2
Oscar Perez	2
Isla Roberts	0
Amelia Walker	2
Ethan Phillips	2
Freya Campbell	2
Benjamin Parker	1
Chloe Evans	0
Arthur Edwards	1
Sienna Collins	0
Thomas Stewart	0
Evie Morris	3
Harry Rogers	6
Zara Cook	0
Samuel Morgan	0

We could then leverage conditional formatting to pull top 10 as at risk

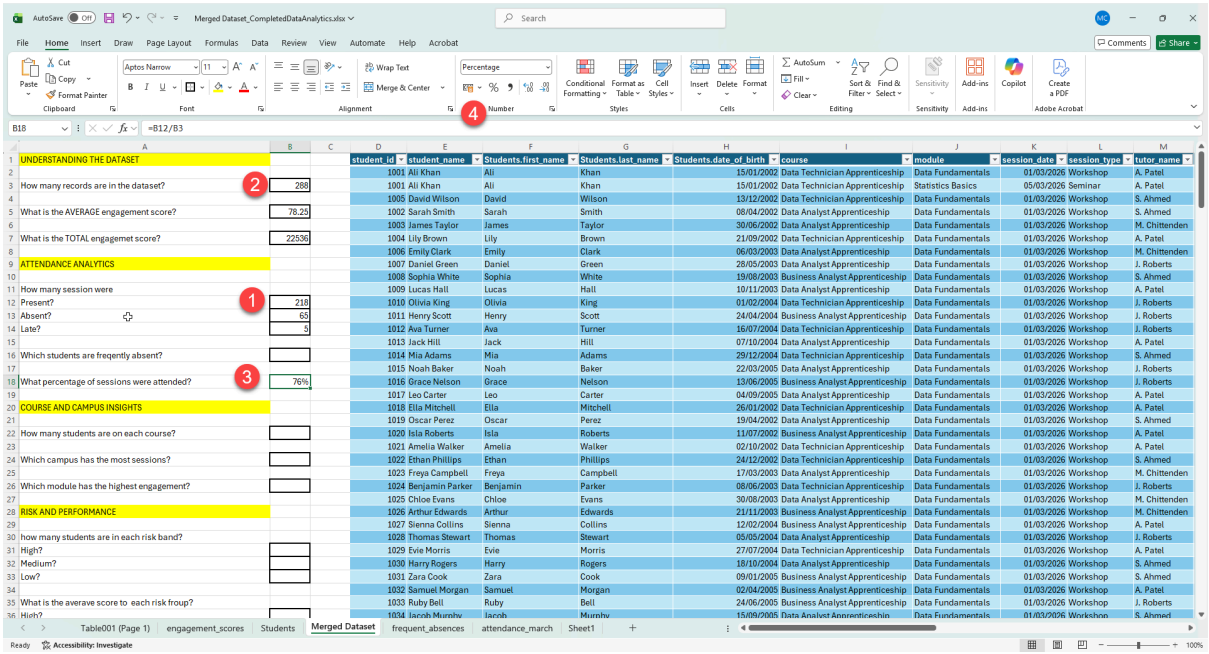


What percentage of sessions were attended?

In comparison to what you have just done this is easy. You already have how many sessions were attended and you previously calculated all sessions so it is one divided by the other.

Step 1:
Divide Cell B12 ÷ Cell B3

Step 2:
Format as percentage



👉 So the answer to this is: **overall attendance rate ~ 76%**

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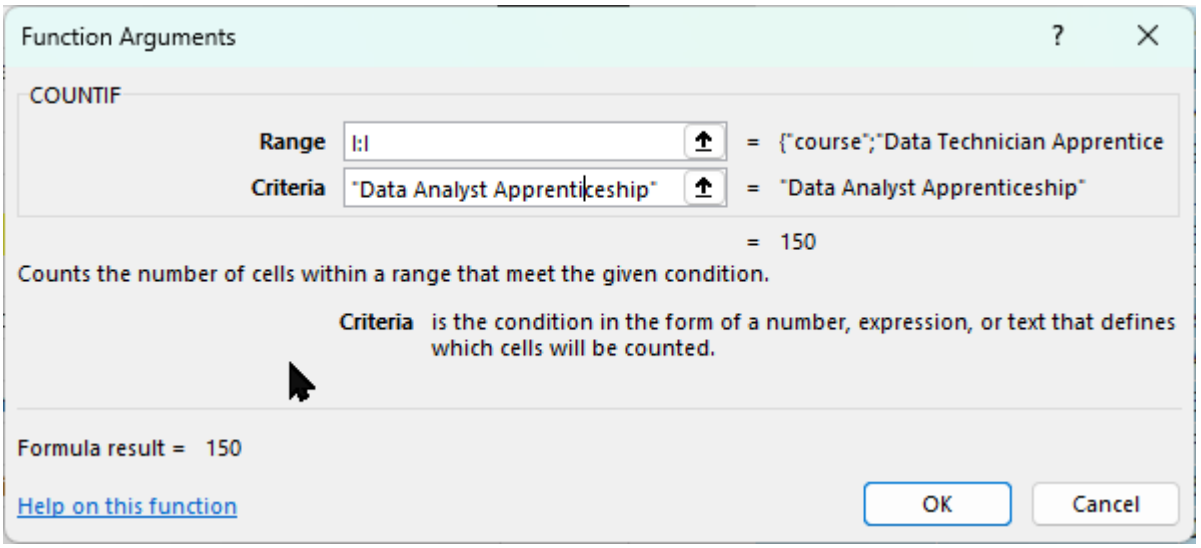
3. Course and Campus Insights

How many students are on each course?

Use:

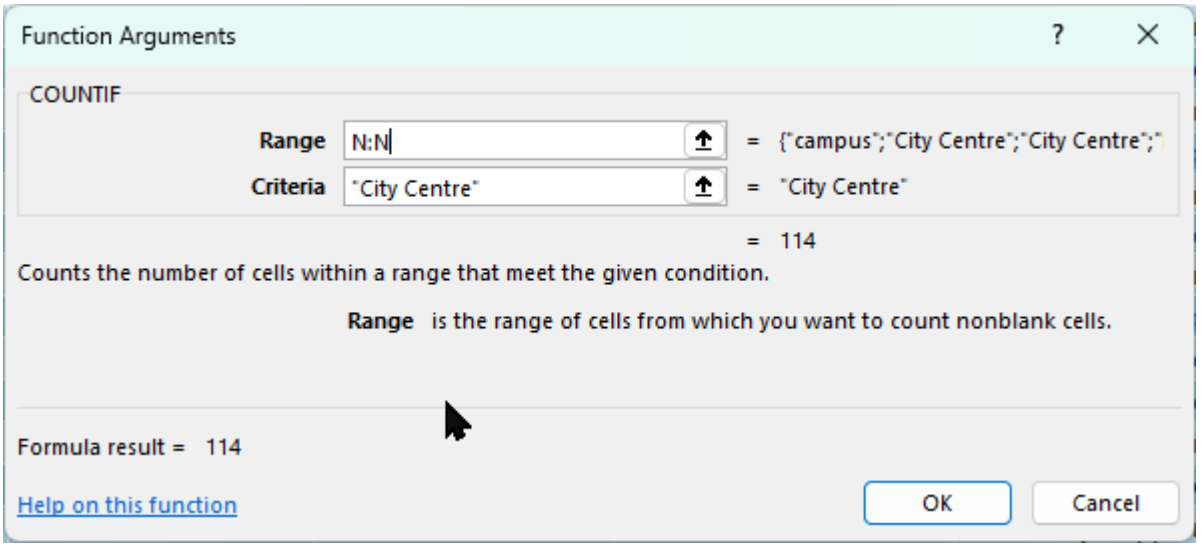
=COUNTIF([Course],"Data Analyst Apprenticeship")



Which campus has the most sessions?

Use:

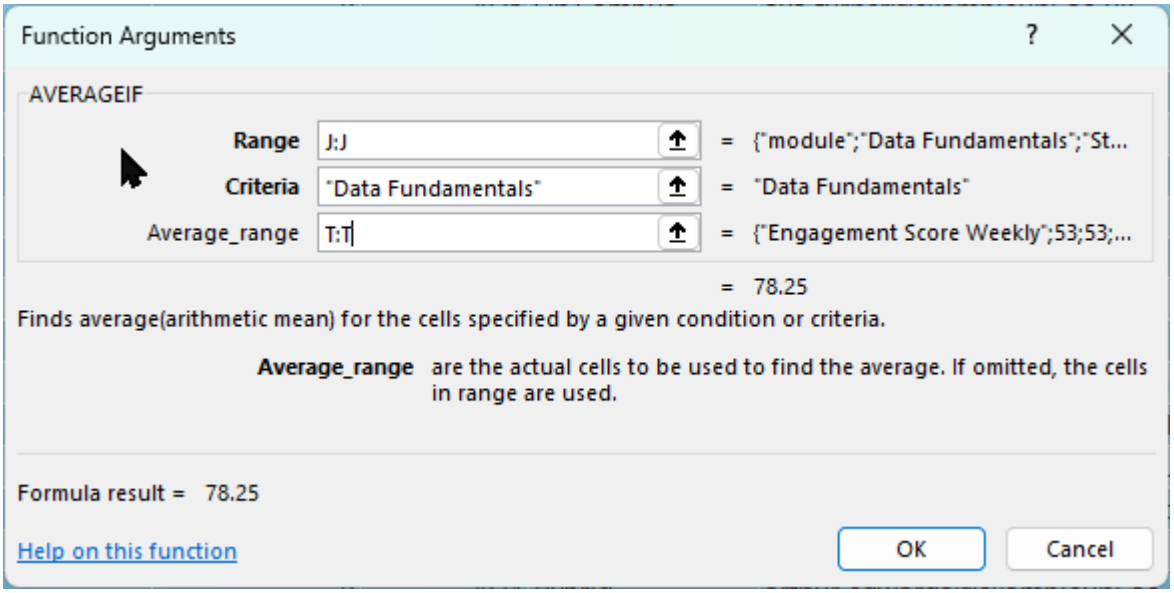
=COUNTIF([Campus],"City Centre")



Which module has the highest engagement?

Use:

=AVERAGEIF([Module],"Data Fundamentals",[Engagement Score Weekly])



4. Risk and Performance

How many students are in each Engagement Risk Band?

Use:

=COUNTIF([Engagement Risk Band],"High")

=COUNTIF([Engagement Risk Band],"Medium")

=COUNTIF([Engagement Risk Band],"Low")

What is the average score for each risk group?

Use:

=AVERAGEIF([Engagement Risk Band],"High",[Engagement Score Weekly])

👉 So the answer to this is: **how engagement differs by risk level**

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5. Withdrawal Insights

How many students have withdrawn?

Use:

=COUNTIF([Status],"Confirmed")

👉 So the answer to this is: **total confirmed withdrawals**

What are the most common withdrawal reasons?

Use:

- Filter → Withdrawal Reason
  OR
- COUNTIF per reason

👉 So the answer to this is: **main causes of withdrawal**

What is the average engagement score of withdrawn students?

Use:

=AVERAGEIF([Status],"Confirmed",[Engagement Score Weekly])

👉 So the answer to this is: **whether low engagement links to withdrawal**

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6. Deeper Analysis (Multiple Conditions)

How many high-risk students were absent?

Use:

=COUNTIFS(

[Attendance Status],"Absent",

[Engagement Risk Band],"High"

)

👉 So the answer to this is: **overlap between risk and poor attendance**

What is the average engagement score for remote students?

Use:

=AVERAGEIF([Delivery Mode],"Remote",[Engagement Score Weekly])

👉 So the answer to this is: **how remote learners are performing**

How many Data Analyst students are Low risk?

Use:

=COUNTIFS(

[Course],"Data Analyst Apprenticeship",

[Engagement Risk Band],"Low"

)

👉 So the answer to this is: **how well that course is performing overall**

Bringing It All Together

Each method supports a different type of question:

- **Filters** → explore the data
- **IF / IFS** → create categories
- **COUNT / COUNTIF(S)** → how many
- **SUM / SUMIF(S)** → how much
- **AVERAGE / AVERAGEIF(S)** → what is typical

Key Concept

Once data has been blended into a single dataset, analysis becomes much more powerful.

You can now:

- Identify **at-risk students**
- Understand **attendance and engagement patterns**
- Compare **courses, campuses, and delivery modes**

Most importantly:
You are no longer just looking at data —
you are **answering real questions and making decisions from it**.